

#This code will be used to demonstrate theorem 6.17 • We will demonstrate that Q2

$$= \frac{((w+z)^3 - y^3) \cdot x \cdot (-w^2 \cdot (z^3 - y^3) + z^2 \cdot (w^3 - y^3))}{y \cdot (w+z)^2 \cdot ((x^3 - w^3) \cdot (z^3 - y^3) + (x^3 - z^3) \cdot (w^3 - y^3))} \text{ is decreasing along the } \gamma \text{ curve; (Part 3 -- calculating the sign of the derivative of } Q2'(\gamma(1), 1))$$

#Define Q2;

$$Q2 := \frac{((w+z)^3 - y^3) \cdot x \cdot (-w^2 \cdot (z^3 - y^3) + z^2 \cdot (w^3 - y^3))}{y \cdot (w+z)^2 \cdot ((x^3 - w^3) \cdot (z^3 - y^3) + (x^3 - z^3) \cdot (w^3 - y^3))} :$$

#Apply the rational parametrization constructed in this chapter 4.

$$\text{parametrization} := \left\{ w = a + \frac{1}{a}, \right.$$

$$z = b + \frac{1}{b},$$

$$y = a - \frac{1}{a} + b - \frac{1}{b},$$

$$x = a - \frac{1}{a} - b + \frac{1}{b} \Big\} :$$

$$q2 := \text{simplify}(\text{expand}(\text{subs}(\text{parametrization}, Q2))) ;$$

$$\begin{aligned} q2 := & \left((a^8 b^4 + 2(2b^5 - b^3)a^7 + (7b^6 - 6b^4 + b^2)a^6 + 2(2b^7 - 6b^5 + b)a^5 + (b^8 - 6b^6 \right. \\ & + 4b^4 + 10b^2 - 1)a^4 + 2(-b^7 + 6b^3 - 2b)a^3 + (b^6 + 10b^4 - 5b^2)a^2 + 2(b^5 \\ & - 2b^3)a - b^4) \left(a^2 b^2 + \frac{1}{3} \right) (a - b)^2 \Big) \Big/ \left(2 \left((b^8 - b^6 + b^4)a^{10} \right. \right. \\ & - \frac{b^2(b^8 - 2b^6 + 8b^4 - 2b^2 + 1)a^8}{2} + \frac{b(b^4 - 4b^2 + 1)(b^2 + 1)^3 a^7}{2} \\ & + \frac{(b^4 + 1)(b^8 + 8b^4 + 1)a^6}{6} - \frac{b(b - 1)^2(b + 1)^2(b^2 + 1)^3 a^5}{2} \\ & \left. \left. - \frac{b^2(b^8 + 2b^6 - 2b^4 + 2b^2 + 1)a^4}{2} - \frac{b^3(b^2 + 1)^3 a^3}{3} + b^6 a^2 - \frac{b^6}{3} \right) (ab + 1) \right) \end{aligned} \quad (1)$$

#Define r1

$$\begin{aligned} r1 := & \left(a^{32} b^{16} + b^{16} + (-4b^{18} + 8b^{16} - 4b^{14})a^{30} + (6b^{20} - 32b^{18} + 44b^{16} - 32b^{14} + 6b^{12})a^{28} \right. \\ & + (72b^{20} - 76b^{18} + 8b^{16} + 692b^{14} + 72b^{12} - 1024b^{10})a^{26} + (b^{32} + 8b^{30} + 20b^{28} \\ & - 120b^{26} - 3426b^{24} - 7288b^{22} - 125524b^{20} - 222104b^{18} - 279672b^{16} - 317080b^{14} \\ & \left. - 157780b^{12} - 32376b^{10} + 1950b^8 + 1672b^6 + 20b^4 + 8b^2 + 1)a^{16} + (-1016b^{26} \right. \end{aligned}$$

$$\begin{aligned}
& - 8440 b^{24} - 21976 b^{22} - 25848 b^{20} - 37428 b^{18} - 31000 b^{16} - 8756 b^{14} + 264 b^{12} - 216 b^{10} \\
& + 8 b^8 + 8 b^6) a^{10} + (-265 b^{24} - 1864 b^{22} + 2192 b^{20} + 4232 b^{18} + 142 b^{16} - 1912 b^{14} \\
& + 144 b^{12} - 72 b^{10} - 9 b^8) a^8 + (-512 b^{22} - 696 b^{20} + 692 b^{18} + 1288 b^{16} - 76 b^{14} \\
& + 72 b^{12}) a^6 + (-250 b^{20} - 288 b^{18} + 44 b^{16} - 32 b^{14} + 6 b^{12}) a^4 + (-4 b^{18} + 8 b^{16} \\
& - 4 b^{14}) a^2 - 2560 \left(b^2 - \frac{1}{5} \right) b^7 (b^2 - 2) (b^2 + 1)^4 a^{23} + 2 b^4 (b^{24} + 36 b^{22} - 2636 b^{20} \\
& - 6620 b^{18} - 27212 b^{16} - 64184 b^{14} - 54798 b^{12} - 25400 b^{10} - 9036 b^8 - 1372 b^6 + 180 b^4 \\
& + 36 b^2 + 1) a^{12} - 4 b^2 (b^{28} + 12 b^{26} - 155 b^{24} + 1018 b^{22} + 4315 b^{20} + 34120 b^{18} + 65514 b^{16} \\
& + 57358 b^{14} + 41514 b^{12} + 19656 b^{10} + 4187 b^8 + 378 b^6 + 37 b^4 + 12 b^2 + 1) a^{14} - 4 b^2 (b^{28} \\
& + 12 b^{26} + 37 b^{24} - 518 b^{22} - 2085 b^{20} + 17992 b^{18} + 38058 b^{16} + 78414 b^{14} + 92394 b^{12} \\
& + 36360 b^{10} + 5211 b^8 - 2118 b^6 - 539 b^4 + 76 b^2 + 1) a^{18} + 2 b^4 (b^{24} + 36 b^{22} + 180 b^{20} \\
& - 2396 b^{18} - 25164 b^{16} - 42552 b^{14} - 90894 b^{12} - 84792 b^{10} - 12876 b^8 - 348 b^6 + 3252 b^4 \\
& + 420 b^2 - 127) a^{20} + 8 \left(b^{20} + b^{18} - 27 b^{16} - 159 b^{14} + \frac{2163}{2} b^{12} - 1827 b^{10} - \frac{1037}{2} b^8 \right. \\
& + 2369 b^6 - 699 b^4 + b^2 + 1 \left. \right) b^6 a^{22} - 9 \left(b^{16} + 8 b^{14} - 16 b^{12} - 72 b^{10} + \frac{3698}{9} b^8 - 328 b^6 \right. \\
& - 528 b^4 + \frac{4168}{9} b^2 + 1 \left. \right) b^8 a^{24} - 1024 b^{15} (b^2 + 1)^4 a^7 - 2560 \left(b^6 - \frac{8}{5} b^4 + \frac{6}{5} b^2 \right. \\
& - 1 \left. \right) b^{11} (b^2 + 1)^4 a^{11} - 1024 \left(b^2 - \frac{1}{2} \right) b^{13} (b^2 + 1)^5 a^9 + 512 b^7 (b^{10} - 19 b^8 + 42 b^6 \\
& - 9 b^4 - 10 b^2 + 1) (b^2 + 1)^4 a^{15} - 1536 \left(b^8 - b^6 + \frac{5}{3} b^4 - \frac{16}{3} b^2 + \frac{5}{3} \right) b^9 (b^2 + 1)^4 a^{13} \\
& + 512 b^7 (b^{10} - 22 b^8 + 81 b^6 - 78 b^4 + 11 b^2 + 1) (b^2 + 1)^4 a^{17} - 3584 b^7 \left(b^8 - \frac{52}{7} b^6 \right. \\
& + \frac{95}{7} b^4 - \frac{43}{7} b^2 + \frac{3}{7} \left. \right) (b^2 + 1)^4 a^{19} + 5632 \left(b^6 - \frac{42}{11} b^4 + \frac{38}{11} b^2 - \frac{5}{11} \right) b^7 (b^2 \\
& + 1)^4 a^{21} \Big):
\end{aligned}$$

#Calculating the derivative of Q2 at b=1

$$d := \text{eval}(\text{diff}(q2, a), b = 1) \cdot \left(-\frac{\text{eval}(\text{diff}(r1, b), b = 1)}{\text{eval}(\text{diff}(r1, a), b = 1)} \right) + \text{eval}(\text{diff}(q2, b), b = 1) :$$

#Locate a range for $\gamma(1)$

$\text{sturm}(\text{sturmseq}(\text{eval}(r1, b = 1), a), a, 2.6, 2.7); \# \text{number of roots in } [2.6, 2.7)$

1

(2)

#Define nd

$nd := \text{numer}(d)$

$$\begin{aligned} nd := & -(a - 1) (27 a^{53} - 27 a^{52} - 192 a^{51} - 60 a^{50} + 291 a^{49} - 363 a^{48} - 5184 a^{47} + 8592 a^{46} \\ & + 39298 a^{45} + 573422 a^{44} - 1028000 a^{43} - 4463672 a^{42} - 11728582 a^{41} + 12872550 a^{40} \\ & + 42134784 a^{39} + 35019856 a^{38} + 57709393 a^{37} + 368530623 a^{36} + 969384096 a^{35} \\ & + 1868298044 a^{34} + 2146658753 a^{33} + 3747524503 a^{32} + 4976666240 a^{31} + 6515419936 a^{30} \\ & + 7240383628 a^{29} + 7560269780 a^{28} + 7320981184 a^{27} + 6366448112 a^{26} + 5770979068 a^{25} \\ & + 4475439108 a^{24} + 3259839744 a^{23} + 1949164832 a^{22} + 1498878157 a^{21} + 787953491 a^{20} \\ & + 488694656 a^{19} + 112973948 a^{18} + 94436917 a^{17} + 24991411 a^{16} - 9251904 a^{15} \\ & + 3779408 a^{14} - 20154542 a^{13} + 541662 a^{12} - 1696544 a^{11} - 2317624 a^{10} + 18090 a^9 \\ & - 153674 a^8 - 67584 a^7 - 6512 a^6 - 4505 a^5 + 105 a^4 + 160 a^3 + 4 a^2 + 7 a + 1) \end{aligned} \quad (3)$$

#Define dd

$dd := \text{denom}(d)$

$$\begin{aligned} dd := & 4 (3 a^{10} - 9 a^8 - 24 a^7 + 10 a^6 - 6 a^4 - 8 a^3 + 3 a^2 - 1)^2 (a + 1)^2 a^3 (a^{28} - 7 a^{24} \\ & - 208 a^{22} + 405 a^{20} + 23552 a^{19} + 1232 a^{18} + 10752 a^{17} - 319075 a^{16} - 48640 a^{15} \\ & - 592416 a^{14} - 26112 a^{13} - 570845 a^{12} + 23040 a^{11} - 398944 a^{10} + 19968 a^9 - 143253 a^8 \\ & + 5632 a^7 - 42000 a^6 - 4608 a^5 + 647 a^4 - 3584 a^3 + 144 a^2 - 65) \end{aligned} \quad (4)$$

#Verify that nd and dd have the same sign in the interval.

$\text{sturm}(\text{sturmseq}(nd, a), a, 2.6, 2.7); \# \text{number of roots in } [2.6, 2.7) ;$

$\text{sturm}(\text{sturmseq}(dd, a), a, 2.6, 2.7); \# \text{number of roots in } [2.6, 2.7)$

0

0

(5)

#Calculate the sign of $Q1'(\gamma(1), 1)$ using $Q1'(3, 1)$

$$\frac{\text{eval}\left(nd, a = \frac{13}{5}\right)}{\text{eval}\left(dd, a = \frac{13}{5}\right)}$$

$$\frac{158244468670558122838933405821826865086530769314992}{1576888028973012906026469483741748527692763899835}$$

(6)